

Xiaotong Shen

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Education

University of Toronto

Honours Bachelor of Science (HSc) in Data Science, Computer Science, Math

Major GPA: 4.0/4.0, cGPA 3.78/4.0

Relevant Coursework: Linear Algebra, Multivariable Calculus with Proof, Software Design, Probability and Statistics with Proofs, Computer Organization, Machine Learning, Data Structures and Analysis, Data Science, Ordinary Differential Equations

Seeking 4–16 Month

Co-op, 2026 Fall

Expected Grad 2028

Experience

ML Research Developer | Java, Typescript

COBWEB

Toronto

August 2025 - Present

- Implemented **XCS** (Accuracy-based Learning Classifier System) within an existing ecological simulation, prioritizing **rule-based interpretability** over black-box **RL** alternatives to support domain researcher understanding.
- Conducted **iterative user testing** with **30+ high school researchers**, evaluating visualization techniques and simulation behavior through structured experimentation to assess pedagogical effectiveness.
- Performed **literature review** on reinforcement learning and ecological modeling frameworks to inform implementation decisions, producing **technical documentation** for future developers.

Outreach Coordinator

SDSS & UTMIST, University of Toronto

Toronto

2025 – 2026

- Led end-to-end planning of a **Python workshop** for **50+ students**, analyzing historical attendance data to inform beginner-focused programming and timing decisions at the start of the academic year.
- Coordinated on planning **5 large-scale events** across SDSS and UTMIST, managing speaker outreach, logistics, and cross-team collaboration to support two of UofT's largest data science and ML communities.
- Managed **sponsor and partner relationships**, conducting cold outreach to companies and speakers and producing marketing content to grow community engagement.

Intern | Python, Excel

Student Success Programs, University of Toronto

Toronto

May 2025 – August 2025

- Extracted and transformed engagement data from **5,000+** student interactions (email open rates, event attendance) using **Python**, loading cleaned datasets into structured formats to identify behavioral trends.
- Authored and presented a comprehensive end-of-term report on content engagement, with **5+ actionable recommendations**, informing future program design, data collection practices, and outreach strategies.
- Designed and published social media content for **@sidneysmithcommons** on Instagram, generating **100,000+ cumulative likes** to increase student awareness of programming.

Section Leader

Stanford University (Code in Place)

Remote

May 2025 – July 2025

- Instructed weekly tutorials for a cohort of **20+** students on core **Python** concepts such as **data structures**, **functions**, and **memory management** as a part of **Stanford University's Code in Place** program.

Projects

AURA: Personality-Aware Emotion Recognition | Python, PyTorch, HuggingFace

UTMIST (2025–2026)

- Surveyed psycholinguistic literature on Big Five personality traits to reformulate the task from continuous regression to **binary classification**, researching evidence that discrete personality prediction is more industry-applicable.
- Fine-tuned a transformer-based classification model in **PyTorch** using **HuggingFace**, adapting RoBERTa on VAD dataset for emotion recognition in conversational agents. (Achieved 0.89 accuracy, improving over baseline by 20 percent)

Web App for Predicting TTC Delays | Javascript, Python

Github (2025)

- Built an end-to-end **machine learning** solution for TTC delay prediction, preprocessed data, trained/evaluated models, and deployed an interactive dashboard built with Streamlit and Plotly for accessible visualization.

Technical Skills

- Languages:** Python, R, SQL, Java, TypeScript, JavaScript, C++
- ML & Data:** PyTorch, TensorFlow, HuggingFace, scikit-learn, pandas, NumPy, Plotly, Streamlit, Jupyter
- Tools & Infrastructure:** Git/GitHub, PostgreSQL, Unix/Bash, REST APIs, OpenAI/Anthropic APIs
- Concepts:** Machine Learning, RL, NLP, Statistical Analysis, Data Visualization, Feature Engineering